WORKED SOLUTION

STEP II 1999, Question 1

Pure • Comparing Large Numbers

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s2-q1

The key idea

These numbers are far too big to compare directly, and even their logarithms are far too big. The way in is to take $\log_{10}$ *twice*. Because x, y and z are each a power of 10, every double logarithm becomes a short sum of 100, x and y, and those are easy to compare. Since $\log_{10}$ is increasing, comparing the double logarithms compares the original numbers.

Solution

Throughout, write log for $\log_{10}$ and

$$L(a) = \log(\log a).$$

The function $t \mapsto \log t$ is strictly increasing, so for numbers $a, b > 10$ (where $\log a, \log b > 1 > 0$) we have $a < b$ exactly when $L(a) < L(b)$. The basic facts we use are

$$\log x = 100, \qquad \log y = x = 10^{100}, \qquad \log z = y = 10^x.$$

Part (i)

Taking $\log_{10}$ of Stirling's approximation with $n = x$,

$$\begin{aligned} \log a_1 &\approx \log\sqrt{2\pi} + \left(x + \tfrac{1}{2}\right)\log x - x\log \mathrm{e} \\ &= 100x - x\log \mathrm{e} + 50 + \log\sqrt{2\pi} \\ &= x\,(100 - \log \mathrm{e}) + 50.4\ldots. \end{aligned}$$

Now $\log \mathrm{e} \approx 0.434$, and the constant $50.4\ldots$ is utterly negligible beside $x = 10^{100}$. So

$$\log a_1 \approx 99.566 \times 10^{100}, \qquad L(a_1) \approx 100 + \log 99.566 \approx 100 + 1.998 = 101.998\ldots \approx 102,$$

as required.

Part (ii)

We compute $\log a_i$ and then $L(a_i)$ for each number.

- $a_2 = x^y$: $\log a_2 = y \log x = 100y$, so $L(a_2) = 2 + \log y = x + 2$.
- $a_3 = y^x$: $\log a_3 = x \log y = x^2 = 10^{200}$, so $L(a_3) = 200$.
- $a_4 = z^x$: $\log a_4 = x \log z = xy$, so $L(a_4) = \log x + \log y = x + 100$.
- $a_5 = \mathrm{e}^{xyz}$: $\log a_5 = xyz \log \mathrm{e}$, so
$$L(a_5) = 100 + x + y + \log(\log \mathrm{e}) \approx y + x + 99.64,$$
since $\log(\log \mathrm{e}) = \log 0.434\ldots \approx -0.36$.
- $a_6 = z^{1/y}$: $\log a_6 = \frac{1}{y} \log z = 1$, so $a_6 = 10$ exactly.
- $a_7 = y^{z/x}$: $\log a_7 = \frac{z}{x} \log y = \frac{z}{x} \cdot x = z$, so $L(a_7) = \log z = y$.

Together with $L(a_1) \approx 102$ from part (i), the double logarithms are

$$\underbrace{0}_{a_6},\ \underbrace{102}_{a_1},\ \underbrace{200}_{a_3},\ \underbrace{x+2}_{a_2},\ \underbrace{x+100}_{a_4},\ \underbrace{y}_{a_7},\ \underbrace{y+x+99.6}_{a_5}.$$

Ask yourself Part (i) only gives an approximation for a_1. Is that good enough to place it?

Yes, and we don't even need Stirling. Each of the x factors of $x!$ is at most x, so $x! \le x^x$ and $\log a_1 \le x \log x = 10^{102}$, giving $L(a_1) \le 102 < 200 = L(a_3)$. And $a_1 = x! > 10 = a_6$ obviously. So $a_6 < a_1 < a_3$ rigorously.

Now we justify the remaining comparisons one at a time.

- $a_3 < a_2$: $200 < 10^{100} + 2$.
- $a_2 < a_4$: $x + 2 < x + 100$.
- $a_4 < a_7$: we need $x + 100 < y = 10^x$. This is true by an enormous margin, because $10^t > t + 100$ for every $t \ge 3$, and here $t = x = 10^{100}$.
- $a_7 < a_5$: $y < y + x + 99.6$, since $x + 99.6 > 0$. (Directly: $\log a_5 = z \cdot xy \log \mathrm{e}$ and $xy \log \mathrm{e} > 1$, so $\log a_5 > z = \log a_7$.)

All the numbers other than a_6 exceed 10, so the monotonicity of L applies, and the order of magnitude is

$$a_6 < a_1 < a_3 < a_2 < a_4 < a_7 < a_5.$$

Checks

For part (i), we compared Stirling's formula with the exact $\log_{10} n!$ (via the log-gamma function) for $n = 10^2, 10^3, 10^5, 10^8$. The double-log formula for a_1 agrees to within 10^{-6},

and for $n = 10^{100}$ it gives 101.998. For part (ii), we repeated the whole calculation with $x = 10^k$ for $k = 2, 3, 4$, where $y = 10^x$ can be held exactly as an integer. The same order $a_6 < a_1 < a_3 < a_2 < a_4 < a_7 < a_5$ comes out each time, so it doesn't hinge on the size of the number 100.

Common slips

- Comparing single logarithms and then getting lost: $\log a_7 = z$ and $\log a_4 = xy$ are still too big to compare by eye. Take logs once more.
- Forgetting the factor $\log \mathrm{e}$ when converting e^{xyz} to base 10. Here it barely matters, but it does shift $L(a_5)$ by about 0.36.
- Missing that a_6 is simply 10, the smallest of all.

Beyond STEP

Iterated logarithms and towers of exponents measure growth in analysis and in computer science. For example, the iterated logarithm $\log^* n$ appears in the running time of some algorithms, and "tower" functions bound results in Ramsey theory. Stirling's formula itself is the first term of an asymptotic series, and proving it carefully is a classic first-year analysis exercise. Abbott's *Understanding Analysis* is a good place to get comfortable with the limits and inequalities involved.

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